|| ॐ नमो भगवते वासुदेवाय नमः ||

18 August 2025 – Monday – 3:45 PM — DLCV

* What is Computer Vision?

→ Computer Vision (CV) is a sub field of AI and deep Learning (DL) that focuses on enabling computers & machines to see, interpret & understand visual data like images, videos or real world scenes, in a way similar to how humans see and percieve the world.

→ The process involves image acquiring, screening, anlyzing, and extracting information by identifying patterns using Deep Learning with Convolutional Neural Networks – CNNs and make decisions for further action.

→ CV projects translate digital visual content into explict descriptions to gather multi-dimensional data. This data is then turned into a computer-readable language to aid the decision making process.

→ Core tasks in CV are :-

- Image classification : Identifying what an image contains
- Object detection : Locating objects in an image with bounding boxes
- Image segmentation : Dividing an image into regions / objects pixel by pixel.
- Face recognition : Identifying individuals based on facial features
- Action recognition : Understanding movements in video.
- Scene understanding : Recognizing the overall context.

* What is OpenCV?

→ OpenCV (Open Source Computer Vision Library) is a free, open-source, cross-platform library of programming functions for computer vision, image processing & ML.

→ It is one of the most widely used libraries in AI & CV because its lightweight, fast & has thousands of optimized algorithms.

→ OpenCV supports a wide variety of programming languages like Python, C++, Java, etc.

→ It can process images & videos to identify objects, faces, tracking motion, feature detection, or even handwriting of a human.

Q) Write a python code to read an image, convert it to HSV, display it & then close the windows. Explain each function used.

Ans)

```
import cv2

img = cv2.imread("image.jpg") # read image

hsv_img = cv2.cvtColor(img, COLOR_BGR2HSV)

cv2.imshow('HSV Image', hsv_img) #display image

cv2.waitKey(0) # wait for a key press
cv2.destroyAllWindows() # destroy all openCV windows
```

→ CV2.imread('image.jpg')

- Reads an image from the file systems.
- Returns the image as NumPy array of pixels.
- By default it loads in BGR (Blue, Green, Red) color format, not RGB.

→ CV2.cvtColor(img, CV2.COLOR_BGR2HSV)

- Converts the color space of the image.
- Here, it transforms BGR → HSV (Hue, Saturation, Value).

→ CV2.imshow('HSV Image', 'hsv_img')

- Opens a window & displays the image.
- First argument: window name (string)
- Second argument: image array to display.

→ CV2.waitKey(0)

- Waits for a key press.
- 0 → wait indefinitely until any key is pressed.

→ CV2.destroyAllWindows()

- Closes all OpenCV windows opened by CV2.imshow()

Q) Write a python code to detect corners in an image using Harris corner Detection algorithm — 5M

Ans)

```
import cv2
import numpy as np

img = cv2.imread('image.jpg') #Read image

gray=cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# convert to float32 (required for Harris)

gray = np.float32 (gray)

#apply harris corner detection

dst = cv2.cornerHarris(gray, blockSize=2,
                       ksize=3, k=0.04)

#dilate corner points for better visibility

  dst= cv2.dilate( dst, None)

#mark corners in red on original image

img[dst >0.01 * dst.max()]=[0,0,255]

#display the result
 cv2. imshow ('Harris Corners', img)
 cv2. waitKey(0)
 cv2. destroyAllWindows()
```

Colour space conversions = changing how colors are represented in an image. We need it in computer vision to make tasks like detection, recognition & segmentation easier and more accurate.

Q) What is colour space conversion? Why do we need it in computer vision? Write a code — 5M

Ans) A colour space is a way of representing colours numerically

In images, colours can be represented in different models such as RGB, HSV, Grayscale, LAB, YCrCb, etc. Colour space conversion means transforming an image from one colour representation to another.

We need colour conversion for different representations for different tasks.

- Grayscale → simplifies data, useful for edge detection, corner detection.
- HSV → better for detecting colours
- LAB | YCrCb → used for skin detection, face recognition.
- Robustness against lighting: RGB is very sensitive to changes in brightness while HSV or LAB are more stable.
- Noise Reduction: Grayscale reduces 3 channels, 1 channel, making computations faster.

☆ CV2.cvtColor(image, code) → Converts bet" colour spaces

- CV2.COLOR_BGR2GRAY → BGR to Grayscale
- CV2.COLOR_BGR2HSV → BGR to HSV
- CV2.COLOR_BGR2LAB → BGR to LAB

Q) Write a Python code to perform rotation of an image using OpenCV.

Ans)

```
import CV2

#Step 1: Load the image
img = CV2.imread('image.jpg')

#Step 2: Get image dimensions (height & width)
(h, w) = img.shape[:2]

#Step 3: Define rotation matrix
#Rotation around center of the image by 45 degree
center = (W//2, h//2)
M = CV2.getRotationMatrix2D(center, 45, 1.0)

#Step 4: Apply the affine transformations (rotation)
rotated = CV2.wrapAffine(img, M, (w, h))

#Step 5: Display results
CV2.imshow('Original Image', img)
CV2.imshow('Rotated Image', rotated)
CV2.waitKey(0)
CV2.destroyAllWindows()
```

- img.shape[:2] → returns dimensions of the image (height, width)

- cv2.getRotationMatrix2D(center, angle, scale)

Creates a 2D rotation matrix
center → point to rotate around.
angle → rotation in degrees. (+ve = counter-clockwise)
scale → zoom factor (1.0 = original size)

- cv2.warpAffine(image, M, (w,h))

Applies rotation matrix M to the image
(w,h) defines the output image size.

☆ Output: Image will be rotated 45° counter clockwise about its center.

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2 September 2025 – Tuesday – 5:20 PM

* Computer Vision

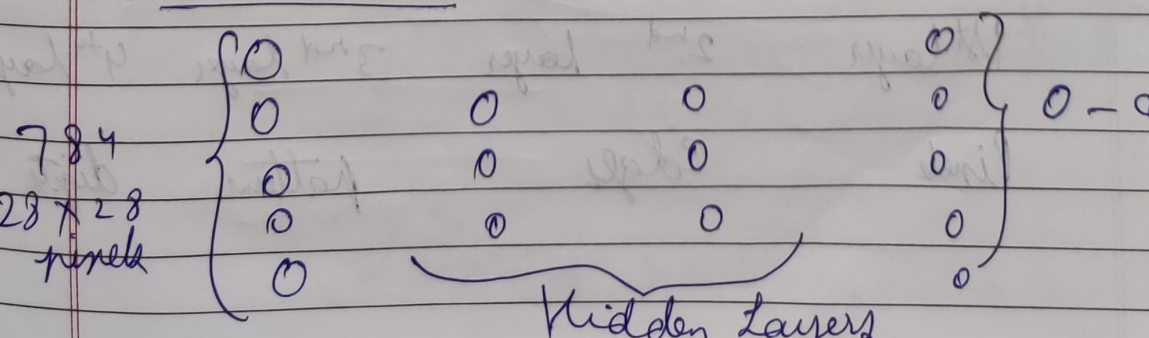
Slide-2 **Introduction:** What PPt is about: This presentation explores how machines see the world (CV) & how AN procss information (MLP)."

- Computer Vision is the ability to give computers a vision i.e. ability to see things and intelligence which is able to process and percieve those visions via neural networks Just like our human brain has neurons which do all the processing for us, neural networks in computer do all the processing. ~~for~~ Its about giving the computers an ability to percieve the world/environment around them just like we humans do with our eyes and understanding of the world.

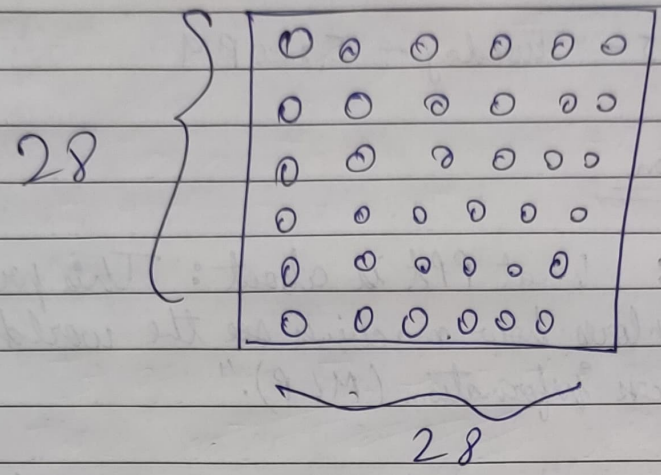
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6 September 2025 – Saturday – 7:25 PM

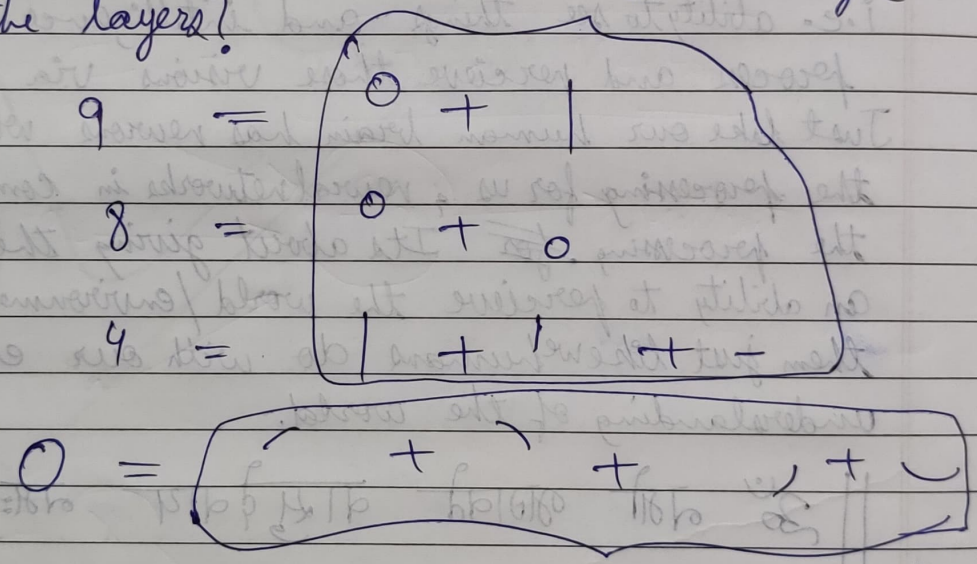
* Neural Networks — DL-1 3B1B



→ Activation is value between 0 to 1



Q) Why the layers?



→ Speech → raw audio → Syllables → words → phrases

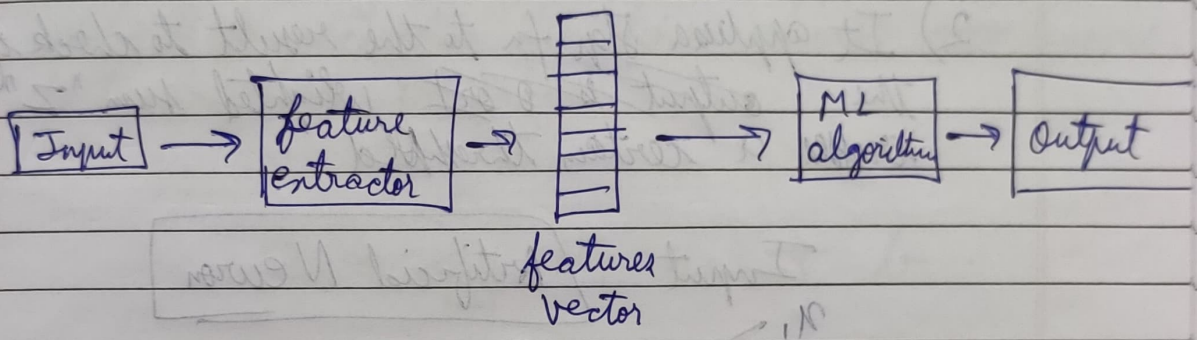
1st layer 2nd layer 3rd layer 4th layer

Pixels Edges patterns digits

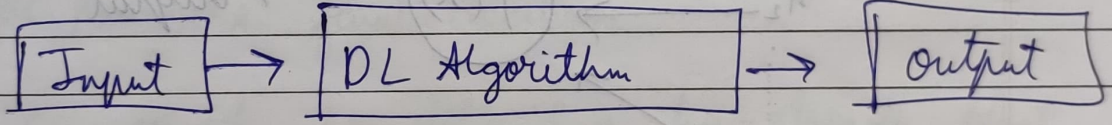
- In traditional ML, we manually extract features that produce a vector of features to be classified by the learning algorithm, whereas in DL, a neural network acts as a both feature extractor & the classifier.

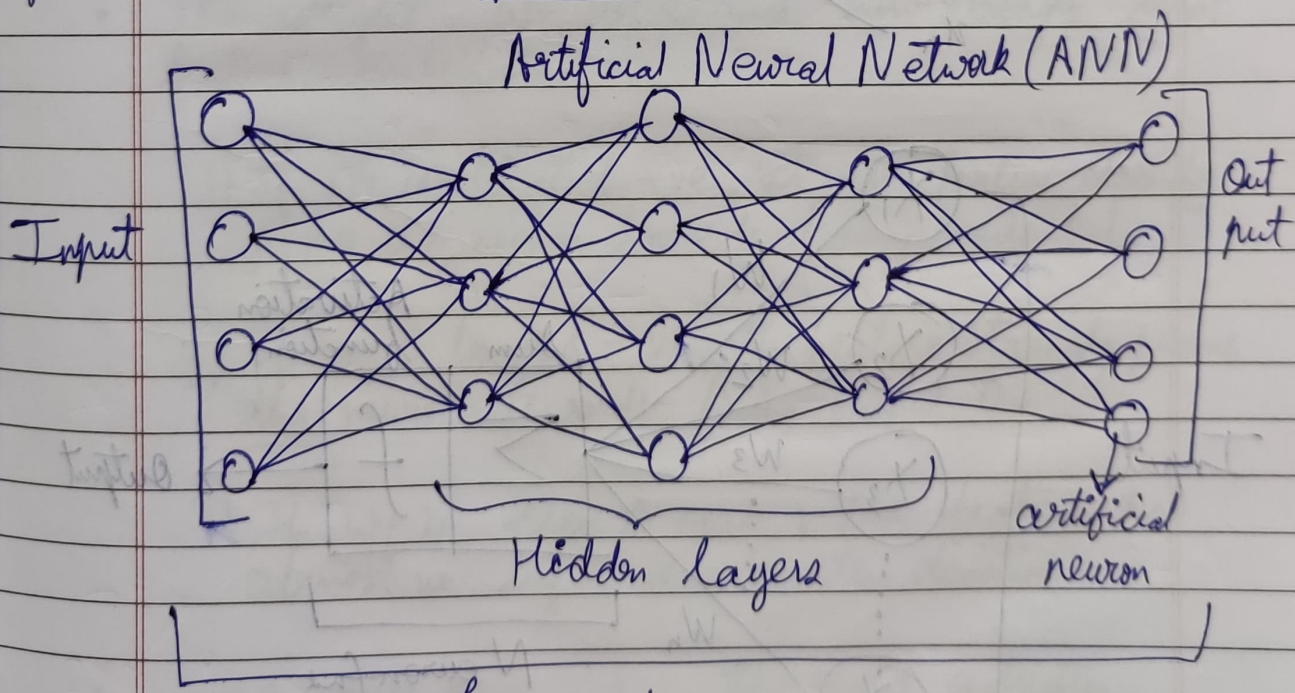
A neural network automatically recognizes patterns & extracts features from the image & classifies them into labels.

ML flow



DL flow



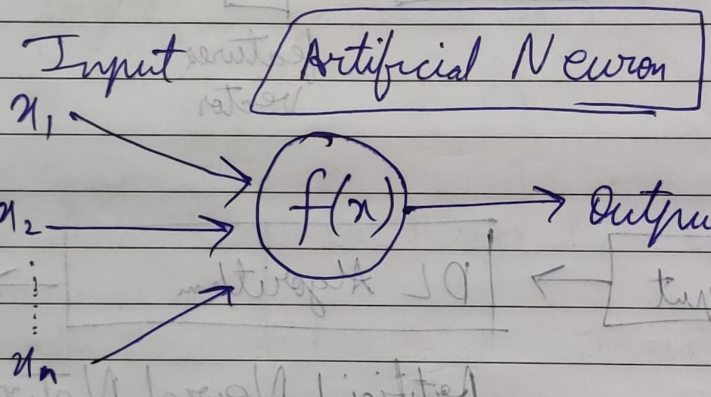


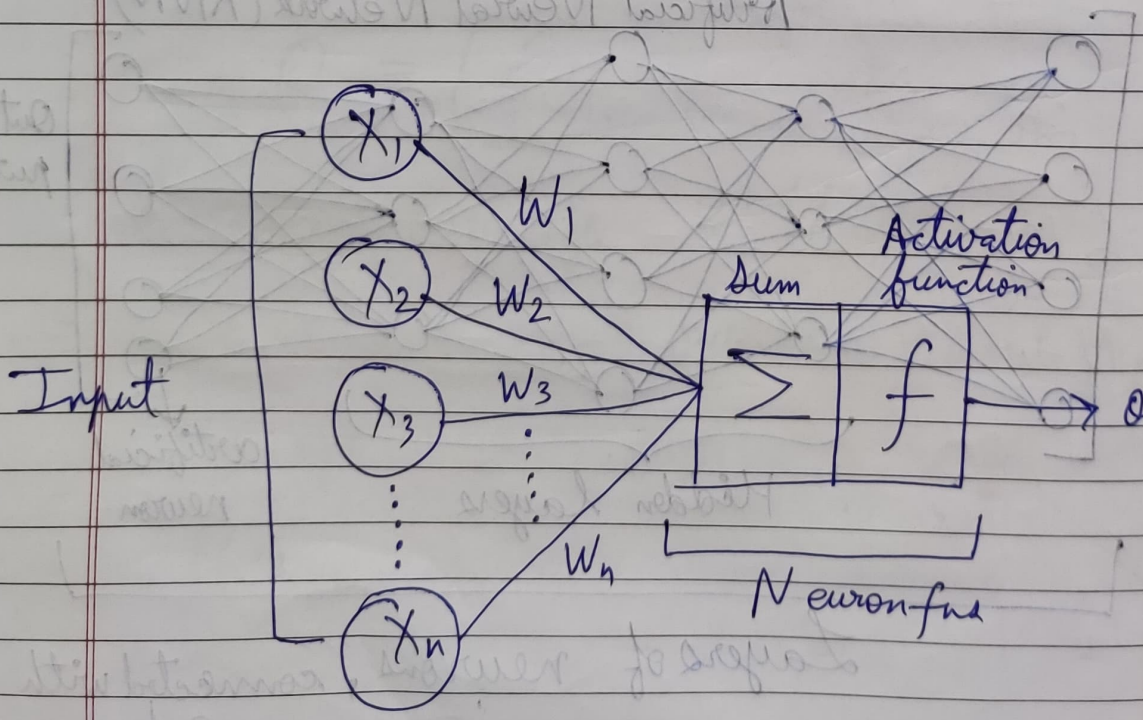
Layers of neurons, connected with edges

- ANN or MLP (Multi Layered Perceptron)
- Artificial neuron performs two functions consecutively

1) it calculates the weighted sum of the 2 inputs to represent the total strength of the input signals

2) It applies a step fn to the result to check if the ~~output is 0 or 1~~ weighted sum "z" crosses a certain threshold





Here

- X_i is the input vector & x_i is the value of it
- W_i is the weight assigned that represents its importance to distinguish between different input datapoints
- Neuron fns are the calculations performed within neuron. It determines the outcome based on the weights & activation functions chosen ~~etc~~ according to the problem & the model.

$$Z = \sum_{i=1}^{n} W_i x_i + b$$

- Each input feature x_i contributes something to the neuron's final decision.
- The weight W_i determines how strongly that feature contributes:

1) If w_i is large & +ve, ~~it~~ the feature strongly pushes the neuron towards firing output 1.

2) If w_i is large & -ve, the feature strongly pushes the neuron towards firing output 0

3) If w_i is close to zero, the feature has almost no influence on the decision.

- b is a bias term like threshold adjuster.

- Weights decidde how much important that particular feature is & how much greater impact will it create in the ~~actfr~~ activation fn in the entire decision making process. Its decides which feature will have greater influence on the outcome of the decision.

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23 September 2025 — Tuesday — 3:53PM

* Image Formation

- In computer vision, image formation refers to the process by which a 3D scene in the real world gets transformed into a 2D image that can be captured, stored & later processed by algorithms.

- Its the pipeline that starts from light interacting with objects then it passes through a camera or sensor then it produces a digital image i.e. a matrix of pixel values.

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29 September 2025 — Monday

Q1) Process of image capture & representation.

Ans1) (1) Image Capture

- Images are captured using devices like camera or scanners.
- A camera lens focus light from the scene onto a sensor (CCD or CMOS).
- The sensors converts light intensity (photons) to electrical signals, representing brightness values.

2) Digitization — Sampling & Quantization

- Sampling — The continuous image is divided into a grid of small elements called pixels.
- Quantization — Each pixels intensity & colour is assigned a discrete numerical value (0–255).

3) Image Representation

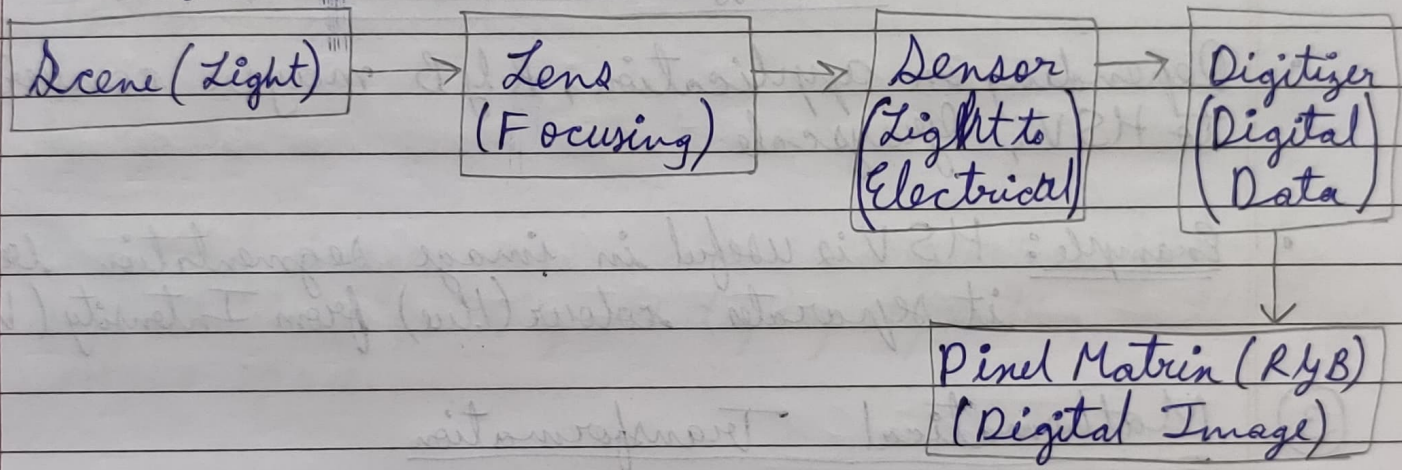
- An image is represented in a computer as a matrix of pixel values.
- A grayscale image, each pixel stores one intensity value.

- In a colour image, each pixel stores three values — R, G, B channels.

4) Storage & Display

- These pixel values are stored in a standard format JPEG, PNG, BMP

- For display, the pixel matrix is mapped back to a screen where each pixel emits corresponding colour & brightness.



Q2) Steps involved in colour space conversion with diagram.

Ans 2) ① Image Capture in RGB

- Most cameras capture images in the RGB colour space.
- Each pixel stores intensity values for R, G & B channels — Red, Green, Blue.

② ~~Identify~~ Identify Target Colour Space

- Depending on application, RGB maybe converted to HSV, Grayscale
- Example: HSV is useful in image segmentation because it separates colour (Hue) from Intensity (Value).

③ Mathematical Transformation

- Conversion uses formulas or transformation matrices.
- Example: RGB → Grayscale

$$Gray = 0.299R + 0.587G + 0.114B$$

RGB → HSV: involves normalizing R, G, B & calculating Hue, Saturation & Value.

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(4) Normalization

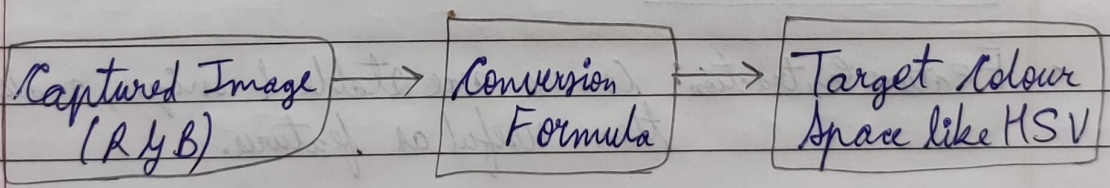
- The resulting values are normalized into a fixed range like 0–255 for 8-bit images.

(5) Representation in New Space

- The image is now represented using the new colour space values.
- For HSV, each pixel stores (H, S, V) instead of (R, G, B).

(6) Application

- Converted images are used in compression, filtering, segmentation, printing or display depending on the target colour space.



Q3) Corner Detection & its importance in CV

Ans3) Corner Detection is a technique in computer vision used to identify points in an image where two edges meet i.e. with strong intensity variation in multiple direction. Corners are also called interest points / keypoints.

* How it works

1) The algorithm analyzes change in pixel intensity in a small neighbourhood.

2) If intensity changes significantly in both X & Y directions, the point is marked as a corner.

3) Common methods are Harris Corner Detector, Shi-Tomasi, FAST, SIFT.

* Importance in CV

- Feature Extraction: Corners are stable & unique, making them useful as features.
- Object Recognition: Helps in identifying & matching objects across images.
- Image Stitching & Panorma: Used in aligning overlapping images.
- Motion Tracking: Corners serve as reliable points to track object movement.

Q4) Explain SIFT, SURF & HoG visual features.

Ans4) *SIFT — Scale Invariant Feature Transform

- It detects & describes local features in images.
- Key properties:

1) Invariant to scale, rotation & partial illumination changes.
2) Detects keypoints by finding extrema in scale-space using Difference of Gaussian (DoG).
3) Each keypoint is described by a 128-dimensional feature vector.

- Uses are object recognition & tracking.

* SURF — Speeded-Up Robust Features

- Faster alternative of SIFT
- Key properties:

1) Uses Hessian matrix approximation for fast keypoint detection.
2) Uses integral images & box filters to speed up computations.
3) Produces 64-dimensional feature descriptors, smaller than SIFT.

- Uses are Real time applications like tracking, panorama.

* **HoG** — Histogram of Oriented Gradients

- Feature descriptor mainly used for object detection like pedestrian detection.
- Key properties:

1) Image is divided into small cells.
2) For each cell, gradients (edge directions) are calculated
3) A histogram of gradient orientations is built for each cell.
4) Descriptors are normalized over blocks to improve robustness.

- Uses are human detection, face recognition, shape analysis.

→ SIFT: Highly accurate but computationally heavy.

→ SURF: Faster, less complex, but slightly less accurate.

→ HoG: Focuses on shape/edges, widely used in detection tasks.

Q5) What is Local Binary Patterns - LBP

Ans 5) LBP is a texture descriptor used in CV. It describes the texture of an image by comparing each pixel with its neighbouring pixels.

* Working Principle

1) For a given pixel, compare its intensity with the neighbouring pixels, usually 3X3 window.

2) If neighbour >= center pixel → assign 1, else 0.

3) The binary values are concatenated into a binary number, then converted into a decimal code.

4) This code represents the local texture pattern.

* Example: Suppose we have a 3X3 pixel block
center pixel = 50

60	55	58
45	50	52
48	49	47

- Compare neighbors with center 50

60 >= 50 → 1, 55 >= 50 → 1, 58 >= 50 → 1, 52 >= 50 → 1
47 >= 50 → 0, 49 >= 50 → 0, 48 >= 50 → 0, 45 >= 50 → 0.

- Binary pattern (clockwise) = 11110000, decimal value = 240
- Thus, LBP value for center pixel is 240.

Q6) What is Scale Space & Scale Selection.

Ans 6) * Scale Space

- In CV, scale space is a framework for representing an image at multiple scales i.e. resolutions.
- It is created by progressively smoothing the image with a Gaussian filter, so both fine details i.e. small scale and coarse structures i.e. large scale can be analyzed.
- Objects or features in an image may appear differently depending on their size or zoom level. Scale space ensures that features can be detected regardless of their size.

* Scale Space

- Scale selection is the process of automatically finding the most appropriate scale at which a feature like edge or corner is best ~~represented~~ detected.
- If you only analyze an image at one fixed scale, you might miss features that are larger or smaller. Scale selection adapts the analysis to the natural size of features.
- Example is SIFT - Scale Invariant Feature Transform uses scale space extrema detection via difference of Gaussian to identify features invariant to scale.

Q7) What is Open CV — 3M

Ans 7) Open CV is a open source library for computer vision, machine learning & image processing.

- It provides tools for image processing, video analysis, face detection, object tracking, machine learning and deep learning integration.
- Open CV supports C/C++, Python, Java
- Used in real-time applications like robotics, surveillance, medical imaging, self driving cars.
- It can process images & videos to identify objects, faces or even the handwritting of a human.

Q8) What is edge detection in C V — 3M

Ans 8) Edge detection is the process of identifying boundaries or discontinuities in an image where there is a sudden change in intensity, colour or texture.

- It highlights object boundaries, shape outlines or texture.
- Reduces amount of data from full image to just boundaries.
- It is the preprocessing step for object recognition, tracking
- Sobel/Prewitt, Canny Edge, Laplacian of Gaussian (LoG) are common edge detection methods.

★ Durga Asthmi ★

30 September 2025 — Tuesday

Q9) CV pipeline

Ans9)

1. Input Data	2. Preprocessing	3. Feature Extraction	4. ML Model
• Images • Videos	Getting the data ready: • Standardize images • Colour transformation • More.....	• Find distinguishing information about image.	Learn from the extracted features to predict & classify objects.

1) Image Acquisition

- Capture image / video using camera, scanner or dataset.
- This is the raw visual data that enters the system.
- Example: a CCTV camera capturing frames.

2) Preprocessing

- Enhance image quality & remove noise.
- Goal is to prepare data for analysis
- Operations include standardizing image sizes, filtering, normalization, histogram equalization, denoising.
- It makes features easier to extract.
- It reduces error for later stages.

3) Feature Extraction

- It is to identify useful patterns or structures.
- Extract distinguishing information such as edges, corners, textures or deep learning features from CNN.
- It converts raw pixels useful numerical descriptor for analysis.
- These features represent important aspects of the image while reducing complexity.

4) ML Model

- The extracted features are fed into a classification model, then the ML model makes predictions based on the features & the information.
- ML or DL models use extracted features to:
  - classify objects
  - detect & localize them
  - classifying objects like bike, car, dog.

→ The CV pipeline involves acquiring input data, preprocessing it, extracting key features & applying an ML model to classify or predict. Each step affects overall accuracy, and improvements can be made by refining any of these stages.

Q10) Geometric Transfor in CV — 3M

Ans10) A geometric transform is an operation that changes the spatial arrangement of pixels in an image.

- It modifies the image's geometry without altering pixel intensity values.
- It maps points (x, y) in the input image to new coordinates (x', y')
- Examples are rotating an image by 90°, shifting it by 10 pixels or correcting distortion.
- Geometric Transform is repositioning or reshaping an image by mapping pixel coordinates.

Q11) Two commonly used colour space conversions — 3M

Ans11) ① RGB ⟶ Grayscale

- Converts a 3-channel of Red, Green, Blue image into a single intensity channel — Gray by removing colour information.

$$Gray = 0.299R + 0.587G + 0.114B$$

- Useful for edge detection, thresholding & feature extraction

② RGB ⟶ HSV (Hue, Saturation, Value)

- Separates colour (Hue), intensity (Value), vibrancy (S)
- Useful in object detection & segmentation where lighting changes.